MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)
THIRD SEMESTER EXAMINATION
(2024 ADMISSION ONWARDS)

MODEL QUESTION PAPER - THEORY

STATISTICS

MG3DSCSTA201-Analytical Tools for Multivariate Analysis

Duration: 1.5 Hours Maximum: 50 Marks

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short answer questions
Answer any seven questions. Each question carries 2 marks.

1. Define orthogonality of vectors and give an example. [K][1]
2. Define basis and dimension. [K][1]
3. Explain the concept of linear independence of vectors with an example. [U][1]
4. Explain orthogonal projection of a vector. [U][1]
5. Write the properties of an idempotent matrix? [A][2]
6. Explain how rank of a matrix is determined using row echelon form? [U][2]
7. Explain the significance of Sylvester's Inequality in matrix rank. [U][2]
8. What is a generalized inverse (G-inverse) of a matrix? [K][2]
9. Define diagonalizable matrix. [K][4]
10. State one application of characteristic roots and vectors in statistics [K][4]

Part B

Short Essay Questions
Answer any four questions. Each question carries 6 marks

11. Define linear independence of vectors and examine the linear independence of the following vectors – (1, 1, -1), (-1, 1, 2), (1, 3, 1). [A][1]

12. Prove that union of two subspaces of a vector space is also a subspace iff one is contained in the other one. [A][1]
13. Briefly explain the multiplication and inverse of a matrix using partitioning. [K][2]
14.. If A and B are 2 square matrices of same order, then show that rank (AB) ≥ rank (A) + rank (B) – n. [K][2]
15. Show that if G is the g-inverse of A and GA = H then (1) H is an idempotent matrix, (2) AH = A, (3) rank(A) = rank(H). [A][2]
16. Briefly explain the spectral decomposition of a square matrix. [U][4]

Part C

Essay questions

Answer any one question. Each question carries 12 marks.

17. Construct an orthonormal set of vectors from the following vectors
X = [1, 1, 0], Y = [0, 1, 1], Z = [1, 0, 1]. [E][1]
18. State and prove Cayley Hamilton theorem. [An][4]